

The two reasons to stop, as research problems: what a correlation inequality would buy, what a certified verification cannot, and where the research is

The user's direction of 2026-09-17: the reasons recorded for stopping are reasons to research. Taking each seriously changes the map, and one earlier statement is corrected: a correlation inequality was called an unblocking, and it is less than that.

What a Griffiths-type inequality would buy. Such an inequality gives $d\langle W(C) \rangle / d\beta_W \geq 0$ for Wilson loops. Consequences: the string tension is nonincreasing in β_W , and the decay rate of any correlator of loop observables with vanishing mean is nonincreasing in β_W . Control therefore transfers **from weaker coupling to stronger coupling only**: knowing a positive mass at β'_W gives it at every $\beta_W \leq \beta'_W$. On the map of the bands note this replaces the three strong-side blocking steps of band B by the single mixing statement H2, and it touches nothing on the weak side, where the problem is. The vacuum-sector gap itself, the 0^{++} channel, is the decay of a truncated correlator whose two terms both increase with β_W , so even its monotonicity does not follow. A correlation inequality is a simplification of H1's target, and no substitute for H1.

Where the standard proofs of such inequalities break. Ginibre's method needs the interaction to lie in a cone of products of single-site functions satisfying the duplicate-variables condition. For the character cone of $SU(2)$ the single-site condition reduces to

$$\iint \prod_{i=1}^n (\chi_i(x) - \chi_i(y)) dx dy = \sum_{S \subseteq \{1..n\}} (-1)^{|S^c|} N_S N_{S^c}, \quad N_S = \dim \text{Hom} \left(1, \bigotimes_{i \in S} V_i \right),$$

which vanishes for odd n , equals $2N_{1234} + 2(\delta_{12}\delta_{34} + \delta_{13}\delta_{24} + \delta_{14}\delta_{23}) \geq 0$ for $n = 4$, and is positive in every case computed at $n = 6$ (six spin $\frac{1}{2}$: 70; six spin 1: 100; four spin $\frac{1}{2}$ and two spin 1: 28; six spin 2: 260). The single-site condition is therefore no obstruction as far as checked. The obstruction is that the plaquette term $\text{Re tr}(U_1 U_2 U_3 U_4)$ is not a nonnegative combination of products of single-link functions: the abelian proof writes $\cos(\theta_1 + \dots)$ through sums and differences of angles, which uses the commutativity of the group, and no analogue exists for a matrix product. To this author's knowledge the second Griffiths inequality is open for $SU(2)$ and $SU(3)$ lattice gauge theory, with neither proof nor counterexample; the centre-based inequalities that do exist are listed in the what-would-unblock note §1.

What a certified verification cannot do. H2 at $\beta_W \simeq 6$ on a box of 216 to 2000 links asks for a certified bound on a supremum over boundary conditions of a total-variation distance between marginals of Gibbs measures on $SU(3)^{|V|}$, an integral over 1700 to 16000 real dimensions. Certified quadrature in that dimension does not exist; transfer-matrix truncations in the character basis have state spaces exponential in the slice; Monte Carlo is not certified. The certified methods that exist are polymer expansions with computer-enumerated small polymers and analytic tails, and they converge only where the expansion does, which for the Wilson measure means β_W of order 0.1 to 1, never 6. So H2 cannot be supplied at the coupling where it is needed, by any method known to the author, and the rigorous reach of the strong side ends at $\beta_W \sim 1$ at best, a factor of six below the crossover. The corresponding decision about the repository's rules is therefore moot.

Where the research is. Both reasons point to the same place. The mass gap for $SU(3)$ is H1: control of the blocking steps from the weak side down to the coupling where the strong side takes over, and the strong side's reach is $\beta_W \sim 1$ in principle and 0.0135 as proved. H1 splits into two sub-targets.

- *H1a, sharpness.* Rigorous small-field renormalization exists (Balaban, Commun. Math. Phys. 95 (1984) 17; 109 (1987) 249; 119 (1988) 243; 122 (1989) 175; Magnen, Rivasseau and Sénéor, Commun. Math. Phys. 155 (1993) 325; Dimock's exposition, Rev. Math. Phys.

25 (2013) 1330010; all metadata level) and holds for g^2 below an unspecified constant. The weak-side analogue of band B is the stretch between the coupling that a rigorous small-field step can handle with explicit constants and $g^2 \simeq 1$, and its width is unknown because no explicit constant has ever been written down for one step.

- *H1b, the physics.* The last steps, in which ξ/a passes from about 10 to about 1; three doublings at one loop.

The programme that follows. Write one blocking step of the Euclidean small-field renormalization for $SU(3)$ with explicit constants, in the form: block-averaged link variables; the fluctuation integral at quadratic order, which is Gaussian with a constraint and whose propagator decays on the scale of the block; the remainder as a polymer gas with an explicit Kotecký–Preiss threshold g_{pert}^2 ; and the bound on the distance of the effective interaction from the Wilson form. The pieces already in hand with constants are the large-field measure bound $e^{-\eta^2/(4g^2)}$ of the lower-bound note, the sharpness of the flow’s growth (instability note), and the lattice truncation (lattice-truncation note). The deliverable of the first step is the number g_{pert}^2 , the weak-side counterpart of the strong side’s 0.0135; the number of doublings between g_{pert}^2 and $g^2 \simeq 1$ is then the honest width of the problem, in place of the crude “ $1/g^2 \gtrsim 10^2$ ” now on the map. Constants explicit; nothing promoted.